Traffic Light Classification using VGG16 Transfer Learning

1. Introduction

Accurate traffic-light colour recognition is important for autonomous vehicles because wrong red or green predictions can lead to unsafe driving behaviour. This project developed a binary image classification system to classify two traffic-light states: Green and Red. I used an expanded dataset containing 2,482 images, with 1,296 Green images and 1,186 Red images. This was an improvement over the original 200-image dataset because it gave the model more examples of traffic-light appearance, lighting variation, background variation and image quality differences.

Transfer learning with VGG16 was used because its pretrained convolutional layers provide strong visual features while only a small task-specific classifier head needs to be trained. Three variations of the model were compared: a frozen VGG16 baseline with data augmentation, an independent improvement using stronger regularisation, and an independent improvement using classification threshold tuning.

2. Dataset Preparation

The final dataset contained 2,482 images organised into Green and Red classes. Images were loaded using OpenCV. Since OpenCV reads images in BGR order, every image was converted to RGB before being passed into the model pipeline. All images were resized to 224 x 224 pixels, matching the standard VGG16 input size. Pixel values were kept in the 0-255 range because the model included the VGG16 preprocess_input function inside the network.

Labels were encoded using LabelEncoder and converted to one-hot format with to_categorical. This allowed the classifier to use a two-neuron softmax output layer with categorical cross-entropy loss. A stratified split was used to keep the Green and Red class distributions approximately balanced in the training, validation and testing subsets. The split used 70% of images for training, 15% for validation and 15% for testing.

Table 1

Stratified dataset split used for training, validation and testing.

Subset	Total images	Green	Red	Purpose
Training	1,737	907	830	Model fitting
Validation	372	194	178	Early stopping and threshold selection
Testing	373	195	178	Evaluation

Data augmentation was applied inside the model so that transformations occurred only during training. The augmentation pipeline included horizontal flipping, small rotation, zooming and contrast adjustment. These operations made the model see varied versions of the same training images and reduced the risk of simply memorising the dataset.

3. Model Design

The baseline model used VGG16 with include_top=False, removing the original ImageNet classification layers. The VGG16 convolutional base was frozen, meaning its pretrained ImageNet weights were not updated during training. A new classification head was added consisting of GlobalAveragePooling2D, Dense (128, ReLU), Dropout (0.25), and a final Dense (2, Softmax) output layer. The two output neurons produced probabilities for the Green and Red classes.

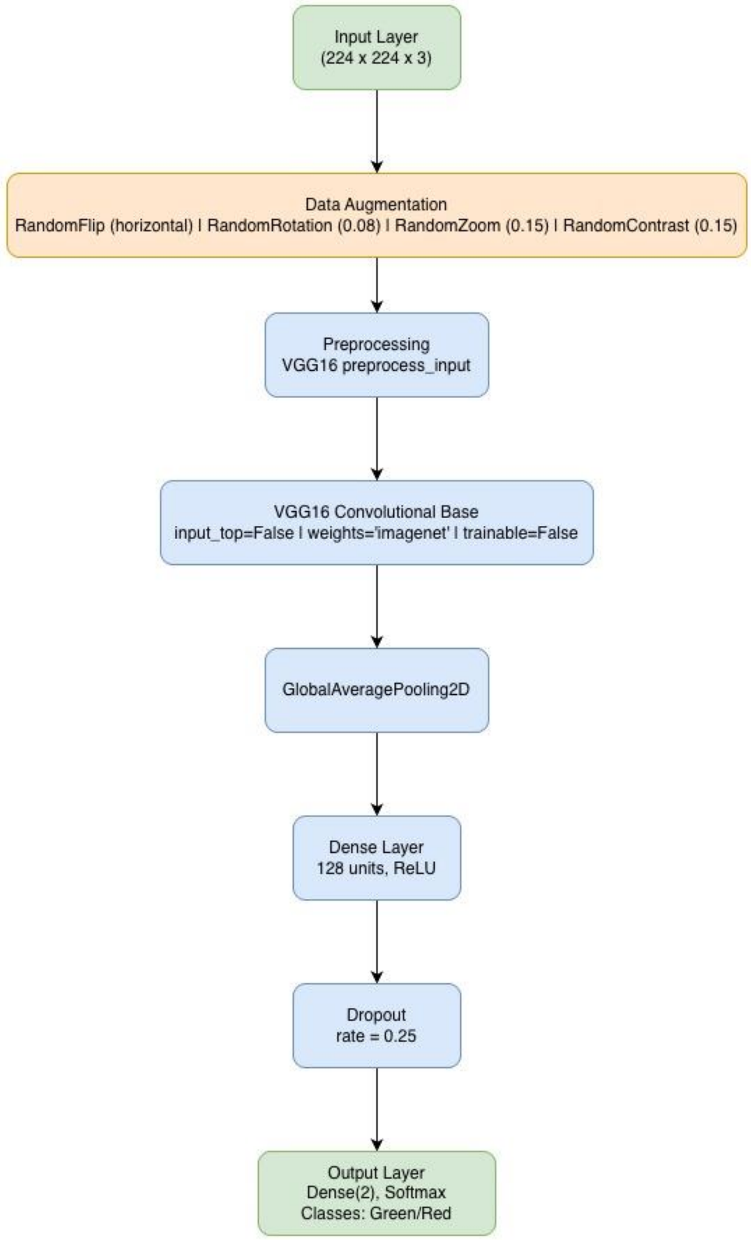


Figure 3.1: Model design for the frozen VGG16 baseline with data augmentation.

The model was trained using the Adam optimiser with a learning rate of 1e-4. Model 1 was the baseline model. Model 2 tested stronger regularisation through L2 regularisation and a higher dropout rate. Model 3 tested threshold tuning by adjusting the Red-class probability threshold using validation-set results.

Table 2

Summary of the baseline VGG16 model architecture.

Layer / operation	Configuration	Purpose
Input	224 x 224 x 3	Receives RGB image
Data augmentation	Flip, rotation, zoom, contrast	Increases training variation
Preprocessing	VGG16 preprocess_input	Matches ImageNet preprocessing
VGG16 base	Frozen, ImageNet weights	Feature extraction
GlobalAveragePooling2D	512 output features	Reduces feature maps
Dense + Dropout	128 ReLU units, dropout 0.25	Classification head
Output	Dense(2), Softmax	Green / Red probabilities

4. Model 1: Frozen VGG16 with Data Augmentation

Model 1 achieved a test accuracy of 97.86% and a test loss of 0.0486. This was a strong baseline result and shows the benefit of using a larger dataset together with transfer learning. The classification report showed high performance for both classes. For the Green class, precision was 0.99, recall was 0.96 and F1-score was 0.98. For the Red class, precision was 0.96, recall was 0.99 and F1-score was 0.98.

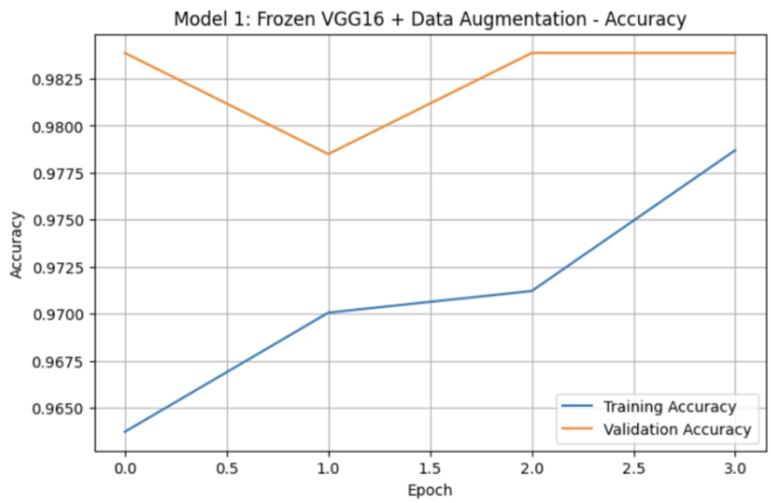
The confusion matrix showed that 188 out of 195 Green images were correctly classified, while 7 Green images were predicted as Red. For the Red class, 177 out of 178 images were correctly classified, with only 1 Red image predicted as Green. The baseline model made 8 total errors on the 373-image test set.

```
Model 1 VGG16 Baseline test accuracy: 0.9786
Model 1 VGG16 Baseline test loss: 0.0486
Classification report for Model 1 VGG16 Baseline:
              precision    recall  f1-score   support

       Green       0.99      0.96      0.98       195
         Red       0.96      0.99      0.98       178

    accuracy                           0.98       373
   macro avg       0.98      0.98      0.98       373
weighted avg       0.98      0.98      0.98       373
```

Figure 4.1: Classification report summary.



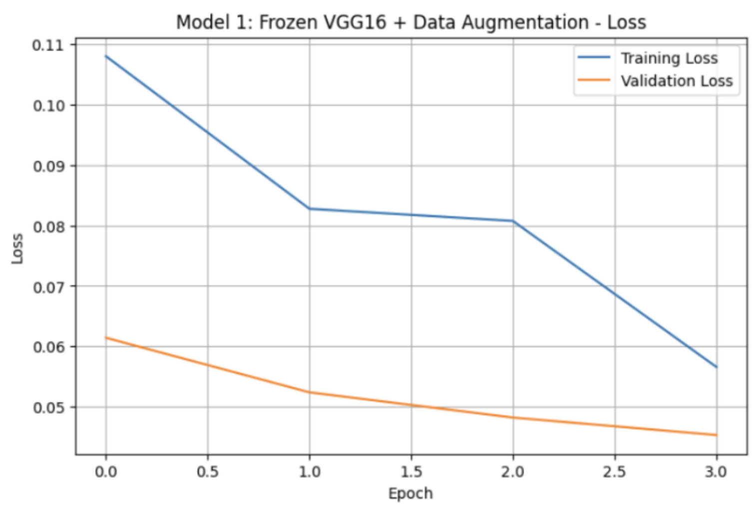


Figure 4.2: Model 1 training and validation accuracy/loss curves.

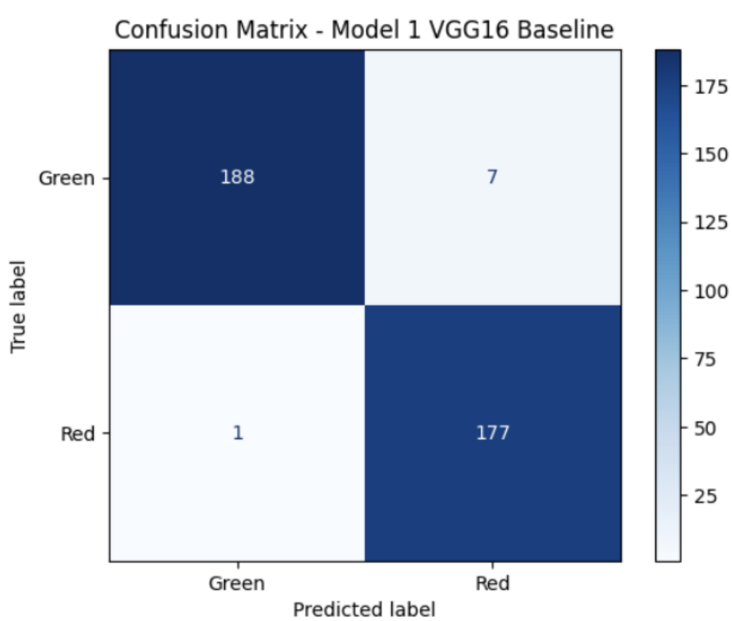


Figure 4.3: Confusion matrix for Model 1.

5. Improvement Method 1: Stronger Regularisation

The first improvement method added L2 regularisation to the dense layer and increased dropout to 0.50 while keeping the VGG16 convolutional base frozen. This was tested as an independent improvement from Model 1. The aim was to reduce overfitting and encourage the classifier head to learn a more general decision boundary.

Model 2 achieved the best overall result, with a test accuracy of 98.93% and a test loss of 0.2232. Green precision was 0.99, recall was 0.98 and F1-score was 0.99. Red precision was 0.98, recall was 0.99 and F1-score was 0.99. Compared with Model 1, the total number of errors decreased from 8 to 4. The confusion matrix showed 192 correctly classified Green images and 177 correctly classified Red images. Only 3 Green images were predicted as Red and only 1 Red image was predicted as Green.

```
Model 2 Regularised VGG16 test accuracy: 0.9893
Model 2 Regularised VGG16 test loss: 0.2232
Classification report for Model 2 Regularised VGG16:
              precision    recall  f1-score   support

       Green       0.99      0.98      0.99       195
         Red       0.98      0.99      0.99       178

    accuracy                           0.99       373
   macro avg       0.99      0.99      0.99       373
weighted avg       0.99      0.99      0.99       373
```

Figure 5.1: Classification report summary.

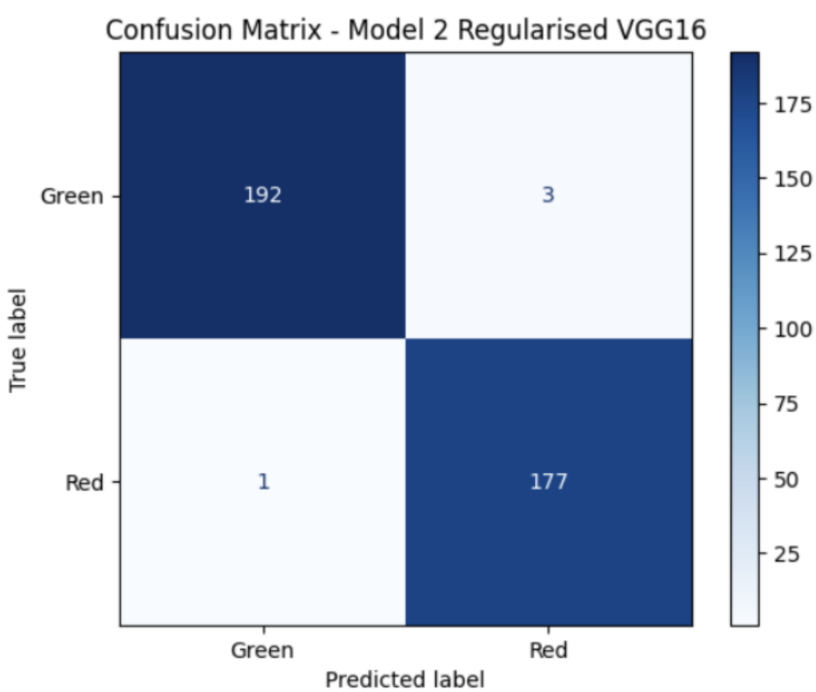


Figure 5.2: Confusion matrix for Model 2.

Figure 5.3: Model 2 training and validation accuracy/loss curves.

This improvement was successful because it improved the baseline test accuracy from 97.86% to 98.93%. It also reduced total errors from 8 to 4, showing that regularisation improved generalisation rather than merely increasing training performance. Model 2 was selected as the best final classifier.

6. Improvement Method 2: Classification Threshold Tuning

The second improvement method was classification threshold tuning which changed the final decision rule used to convert the baseline model probabilities into class labels. Different Red-class probability thresholds were tested using the validation set, and the threshold with the best validation macro F1-score was selected. This method was applied to Model 1 probabilities.

The best Red threshold was 0.65. At this threshold, validation accuracy was 0.9866 and validation macro F1-score was 0.9865. On the test set, threshold tuning achieved 98.66% accuracy, macro precision of 0.9863, macro recall of 0.9869 and macro F1-score of 0.9866. The confusion matrix showed 191 correctly classified Green images and 177 correctly classified Red images. There were 4 Green-to-Red errors and 1 Red-to-Green error, giving 5 total errors. This improved Model 1, but it did not outperform Model 2.

Table 3

Threshold tuning summary.

Metric	Value
Best Red threshold	0.65
Validation accuracy	0.9866
Validation macro F1	0.9865
Test accuracy	98.66%
Test macro F1-score	0.9866

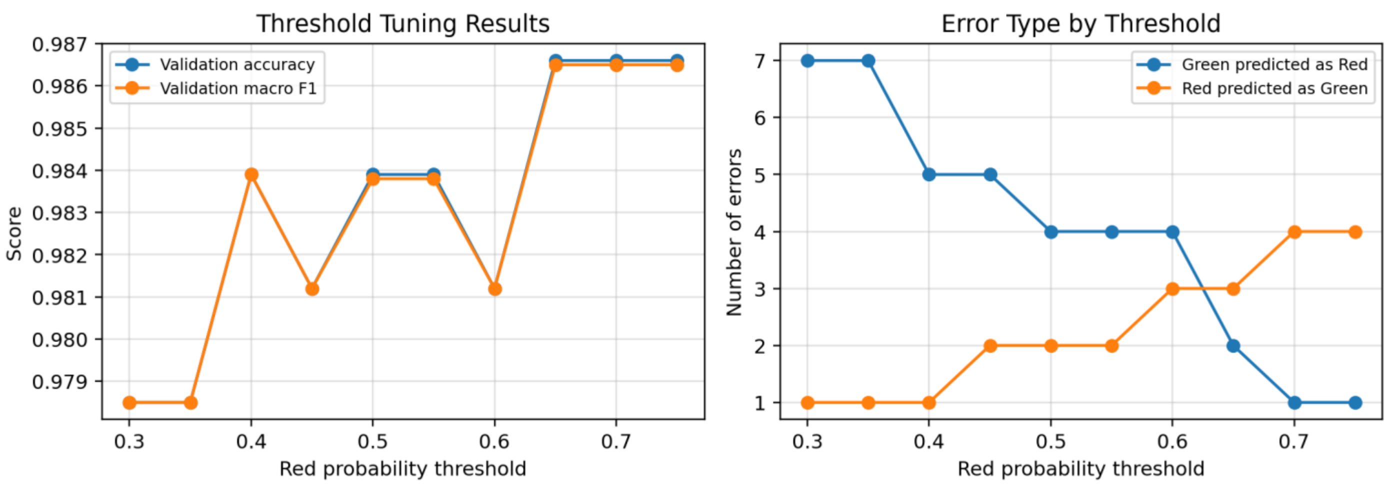


Figure 6.1: Threshold tuning validation results and error-type trade-off.

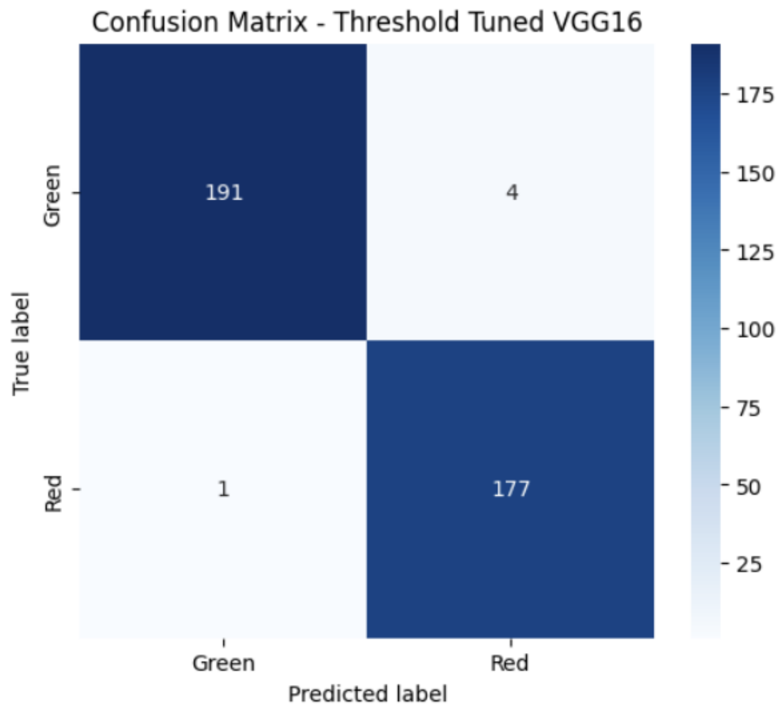


Figure 6.2: Confusion matrix after threshold tuning.

The error-type plot showed the trade-off created by threshold tuning. Increasing the Red threshold reduced Green images being predicted as Red, but it also increased the risk of Red images being predicted as Green. This matters because Red-to-Green errors are more safety-critical in autonomous driving. The selected threshold balanced these two error types well and improved the baseline, but Model 2 remained the strongest result overall.

7. Model Comparison

Table 4

Final comparison of the baseline and independent improvement methods.

Model	Method	Test accuracy	Errors	Main finding
Model 1	Frozen VGG16 + data augmentation	97.86%	8	Strong baseline
Model 2	Stronger regularisation	98.93%	4	Best final model
Model 3	Threshold tuning on baseline	98.66%	5	Improved Model 1, but below Model 2

The comparison shows that both improvement methods improved on Model 1 independently. Stronger regularisation improved accuracy from 97.86% to 98.93%, while threshold tuning improved accuracy from 97.86% to 98.66%. Model 2 was selected as the best final classifier because it achieved the highest test accuracy and the lowest number of test errors.

8. Five Unseen Test Image Predictions

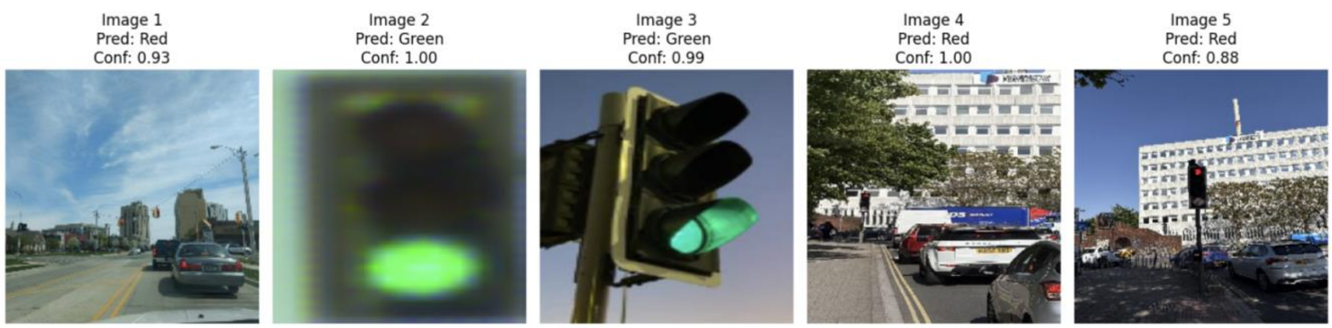


Figure 8.1: Five unseen test images results

Table 7

Five selected unseen test-image predictions using the final selected model.

Image	Actual label	Predicted label	Confidence
1	Red	Red	0.93
2	Green	Green	1.00
3	Green	Green	0.99
4	Red	Red	1.00
5	Red	Red	0.88

The five sample predictions demonstrate that the final model correctly classified five out of the five displayed test images. Three Red images were correctly predicted as Red, and two Green images were correctly predicted as Green.

9. Conclusion

Both improvement methods improved the VGG16 baseline. Accuracy increased from 97.86% to 98.93% with stronger regularisation and 98.66% with threshold tuning. Green recall improved from 0.96 to 0.98, while Red recall remained high at approximately 0.99. The final selected classifier was Model 2, the regularised VGG16 model. It achieved the best test accuracy and the lowest error count.